

PCOS Care AI

User Manual

AI-based preliminary PCOS risk assessment and personalised, non-diagnostic care coordination between patients, doctors and counselors.

■ Not a medical device. PCOS Care AI provides an AI-generated preliminary risk estimate only. It does not diagnose PCOS and does not replace a qualified healthcare professional.

Contents

1. Introduction
2. Getting Started
3. For Patients
4. For Doctors
5. For Counselors
6. For Administrators
7. Frequently Asked Questions
8. Support & Disclaimer

1. Introduction

PCOS Care AI is a web application that helps people understand their risk of Polycystic Ovary Syndrome (PCOS) based on common symptoms and lifestyle factors. A Random Forest machine-learning model analyses the answers and returns a preliminary risk level, along with general wellness guidance.

Beyond the risk assessment, the app connects patients with a doctor or counselor: high-risk patients can be routed to immediate care, providers can review assessments and blood/lab reports, and upload prescriptions.

Note: This tool is not a medical device. It does not diagnose PCOS. Always consult a qualified healthcare professional for diagnosis and treatment.

Who uses this app?

- **Patients** – take assessments, track results, choose a care provider, receive prescriptions.
- **Doctors** – review every patient's assessments and reports, upload prescriptions, toggle their availability.
- **Counselors** – support the doctor for high-risk patients who choose them, when the doctor is unavailable.
- **Administrators** – monitor model performance and overall usage statistics.

2. Getting Started

Creating a patient account

- 1 Open the app and click **Create an account** on the Login page (or go to **/register**).
- 2 Enter your name, email address and a password (minimum 6 characters).
- 3 You're automatically logged in and taken to your Patient Dashboard.

Logging in

All roles – patients, doctors, counselors and administrators – use the same **Login** page with an email and password. After logging in, you're taken to the dashboard for your role.

Demo accounts

The app ships with ready-to-use demo accounts for evaluation:

Role	Email	Password
Doctor	doctor@pcos.ai	doctor123
Counselor	ananya.counselor@pcos.ai	counselor123
Counselor	priya.counselor@pcos.ai	counselor123
Counselor	fatima.counselor@pcos.ai	counselor123

Admin	admin@pcos.ai	admin123
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Note: Patients should register their own account rather than using a shared demo login.

Forgot your password?

Click **Forgot password?** on the Login page and enter your email. A password-reset link is generated for your account (in this demo build, administrators can find it in the server console since no email provider is configured). Opening the link lets you set a new password; the link expires after 30 minutes and can only be used once.

3. For Patients

3.1 Taking an assessment

From the Home page or navigation bar, click **Assessment**. Fill in your basic information (name, age, cycle length, weight, height) and answer the symptom & lifestyle questions honestly – your BMI is calculated automatically as you type. Submit the form to get your result.

Note: You can take an assessment without an account, but signing in lets your results, prescriptions and reports be saved to your dashboard.

3.2 Understanding your result

Your result page shows:

- A **risk gauge** (Low, Moderate or High) with the model's estimated probability.
- Your calculated **BMI**.
- **Personalised recommendations** grouped by category (Nutrition, Activity, Weight, Monitoring, Wellbeing, Next steps).
- A disclaimer reminding you this is not a medical diagnosis.

You can download a PDF copy of your result at any time using the **Download PDF report** button.

3.3 High-risk alert: choosing a doctor or counselor

If your result is **High** risk, a popup immediately recommends contacting a doctor, showing their name, specialty, email and phone number.

- If the doctor is currently **available**, only the doctor's contact card is shown.
- If the doctor is **unavailable**, the popup instead lists all counselors (with their contact details) so you can pick one.
- If you're logged in as a patient, clicking **Select** saves that person as your care provider. Guests just see the contact details.

Note: Counselors are only ever offered for High-risk results. For Moderate or Low risk, your care stays with the doctor – no counselor option is shown, even if the doctor is unavailable.

3.4 Your Patient Dashboard

Visit **My Dashboard** (after logging in) to manage everything in one place:

Your care provider

New patients are assigned to the doctor by default. If your latest assessment is High risk and the doctor is unavailable, you can switch to one of the counselors here – you can change your choice at any time.

Past assessments

A table of every assessment you've taken, with date, age, BMI, probability and risk level.

Prescriptions from your provider

Prescriptions uploaded for you by your doctor or counselor appear here, newest first, with a download button. A doctor's prescription is always marked **Current**; if a counselor had uploaded one earlier for the same visit, it's marked **Superseded by your doctor** once the doctor's prescription arrives – both stay visible for your records.

Blood / lab reports

If your latest assessment is **High risk**, you can upload a blood or lab report (PDF or image, up to 10 MB) for your doctor or counselor to review before they prescribe anything. Previously uploaded reports remain listed and downloadable even if your risk level later changes.

3.5 Logging out

Use the **Logout** button in the navigation bar at any time.

4. For Doctors

Log in with your doctor account to reach the **Doctor dashboard**, which shows **every** patient's assessments – not just the ones who selected you – regardless of your own availability status.

4.1 Availability toggle

The doctor is the only role with an availability switch, shown at the top of the dashboard. Toggle it to **Unavailable** when you can't take on new patients – this is what triggers counselors being offered to patients with a High-risk result.

Note: Toggling yourself unavailable never restricts what you yourself can see or do – you always retain full access to every patient's records.

4.2 Reviewing assessments

The assessments table can be filtered by risk level (All / High / Moderate / Low). For each row you can:

- **Report** – download a full PDF report for that specific assessment.
- **Rx** – upload a prescription for that patient.
- **Labs** – view and download any blood/lab reports that patient has uploaded.

4.3 Uploading prescriptions

Click **Rx** on any row, attach a PDF or image file (up to 10 MB) and an optional note, then submit. As a doctor, you can upload a prescription for **any** patient, regardless of whether they selected you or a counselor as their provider.

Note: When you upload a prescription, any earlier prescription a counselor had uploaded for that same patient is automatically marked as superseded – your prescription always takes priority, without deleting the counselor's record.

4.4 Viewing blood / lab reports

Click **Labs** on any row to see every report that patient has uploaded, with a download link for each – available to you for any patient, at any risk level.

5. For Counselors

Log in with your counselor account to reach the **Counselor dashboard**. Unlike the doctor, a counselor's access is scoped: you only see assessments for patients who have specifically selected you as their provider, and only for their **High-risk** assessments.

Note: Counselors have no availability toggle – you're simply offered to patients whenever the doctor is marked unavailable.

5.1 Reviewing your patients

Your dashboard lists only the patients who chose you. If no one has selected you yet, you'll see "No patients have selected you yet."

5.2 Uploading prescriptions

The **Rx** button is only enabled for rows where that patient's assessment is High risk. For Moderate or Low risk assessments, prescribing stays with the doctor and the button is disabled with an explanatory tooltip.

5.3 Viewing blood / lab reports

The **Labs** button follows the same rule: you can view a patient's uploaded reports only while their latest assessment is High risk.

5.4 If the doctor uploads a prescription

If the doctor later uploads their own prescription for one of your patients, it takes priority. Your earlier prescription for that patient is marked "Superseded" in the patient's view but is never deleted.

6. For Administrators

Log in with the admin account to reach the **Admin dashboard**, which gives a full picture of the model's performance and the app's usage.

6.1 Model performance

Evaluated on a held-out 20% test split of the training data:

- **Accuracy** – share of test cases classified correctly overall.
- **Precision** – of predicted PCOS-risk cases, how many were actually right.
- **Recall** – of true PCOS-risk cases, how many the model caught.
- **F1-score** – the balance between precision and recall.
- **ROC-AUC** – how well the model ranks risk overall (1.0 = perfect).
- A **confusion matrix** and **feature importance** chart, showing which inputs most influence the model.

6.2 Usage statistics

- Total assessments submitted, and the average predicted risk probability.
- The breakdown of Low / Moderate / High risk results.
- The 10 most recent assessments across the whole app.

6.3 Retraining the model

Click **Retrain model** to re-run training and refresh the metrics shown on this page. This is mainly useful after changing the underlying training data or configuration.

7. Frequently Asked Questions

Q: Is this a medical diagnosis?

A: No. PCOS Care AI gives an AI-generated preliminary risk estimate only. Always consult a qualified healthcare professional for diagnosis and treatment.

Q: Can I take an assessment without creating an account?

A: Yes, but your result won't be saved anywhere, and you won't be able to select a care provider, receive prescriptions, or upload blood reports.

Q: Why can't I select a counselor?

A: Counselors are only offered when your latest assessment is High risk and the doctor is currently marked unavailable. For Moderate/Low risk, only the doctor is involved.

Q: Why is my Rx / Labs button disabled (as a counselor)?

A: Counselors can only act on patients who selected them and whose latest assessment is High risk. The button's tooltip explains which condition isn't met.

Q: I forgot my password – what do I do?

A: Use “Forgot password?” on the Login page. A reset link is generated for your account and expires after 30 minutes.

Q: Can I change my care provider later?

A: Yes – from your Patient Dashboard's “Your care provider” section, whenever a counselor option is available to you.

Q: What happens to a counselor's prescription once the doctor uploads one?

A: It's marked “Superseded by your doctor” but stays visible in your prescription history – nothing is deleted.

8. Support & Disclaimer

PCOS Care AI · Preliminary risk-assessment & awareness tool.

Note: Not a medical device. PCOS Care AI provides an AI-generated preliminary risk estimate only. It does not diagnose PCOS and does not replace a qualified healthcare professional. Always consult a doctor or gynaecologist for proper clinical evaluation, especially if your result indicates elevated risk.

For technical issues or questions about this application, please contact your system administrator.